Hilbert-Valued Characteristic Domination for General Local Martingales in L^2

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2026-06-16

Status

This packet gives a candidate partial result, likely valid, for the open general-local-martingale problem stated in Remark 12.10 of Yaroslavtsev's paper *Local characteristics and tangency of vector-valued martingales* [1]. The result proves the desired strong estimate for general Hilbert-valued local martingales at the exponent $p = 2$. It is not a solution of the full UMD problem for all exponents.

Source Problem

The source is arXiv:1907.11588, PDF page 105, Remark 12.10. The source crop below states the open general-local-martingale issue and its discrete independent-increment reduction.

(12.11), (12.14), (2.9), and the fact that M and N are quasi-left continuous. □

Remark 12.10. It is not known whether Theorem 12.8 holds for general local martingales. By Theorem 3.39 and by Proposition B.1 the main issue here is in proving a similar statement for discrete martingales with independent increments. Let us state this problem here as open. Let X be a Banach space, $1 \leq p < \infty$. Let $(\xi_n)_{n\geq 1}$ and $(\eta_n)_{n\geq 1}$ be X-valued mean-zero independent random variables such that for any Borel $B \in X \setminus \{0\}$

$$\sum_n \mathbb{P}(\xi_n \in B) \leq \sum_n \mathbb{P}(\eta_n \in B).$$

Does there exists a constant C (perhaps depending on p and X) such that

$$\mathbb{E}\Big\|\sum_n \xi_n\Big\|^p \leq C\mathbb{E}\Big\|\sum_n \eta_n\Big\|^p?$$

The next page adds that, by symmetrization, one can assume the variables are symmetric, and that even this case was not known to the author.

By a standard symmetrization trick [66, Lemma 6.3] one can assume that ξ_n's and η_n's are symmetric. But even the symmetric case is not known for the author.

In the notation of the source, a local martingale N is characteristically dominated by M if the initial weak coordinates are dominated, the continuous covariation bilinear form of N is dominated by that of M, and the compensator ν^N of the jump measure is dominated by ν^M. The question is whether the quasi-left-continuous estimate from Theorem 12.8 remains true for general local martingales.

Idea of the Proof

At $p = 2$, Hilbert spaces retain exactly the information that characteristic domination records. For a Hilbert-valued square-integrable martingale, the terminal second moment is the initial square norm plus the trace of the continuous quadratic covariation plus the integral of $\|x\|^2$ against the jump compensator. Characteristic domination decreases each of these terms. Thus the terminal L^2 estimate has constant 1. The strong maximal estimate follows by Doob's L^2 inequality.

Hilbert L^2 Subcase of the General Problem

Lemma 1 (Hilbert square identity)**.** *Let H be a separable Hilbert space and let M be an H-valued square-integrable martingale on $[0,T]$. Let $\langle\langle M^c\rangle\rangle$ denote the predictable covariation bilinear form of the continuous part, and let ν^M be the compensator of the jump measure. Then*

$$\mathbb{E}\,\|M_T\|^2 = \mathbb{E}\,\|M_0\|^2 + \mathbb{E}\ \mathrm{tr}\langle\langle M^c\rangle\rangle_T + \mathbb{E}\int_{[0,T]\times H} \|x\|^2\ d\nu^M(t,x),$$

with the value $+\infty$ allowed.

Proof. For finite-dimensional H, this is the usual coordinatewise martingale isometry applied to an orthonormal basis: the continuous part contributes the trace of its predictable covariation and the purely discontinuous part contributes the compensator of $\sum_{s\le T}\|\Delta M_s\|^2$.

For separable H, let P_m be the orthogonal projection onto the span of the first m vectors of an orthonormal basis. Apply the finite-dimensional identity to $P_m M$. Then let $m\to\infty$ and use monotone convergence for $\|P_m M_T\|^2$, for the finite-dimensional traces, and for $\int \|P_m x\|^2\ d\nu^M(t,x)$. □

Theorem 1. *Let H be a separable Hilbert space. Let M and N be H-valued local martingales such that N is characteristically dominated by M in the sense of Yaroslavtsev. Then for every bounded stopping time τ,*

$$\mathbb{E}\,\|N_\tau\|^2 \le \mathbb{E}\,\|M_\tau\|^2\,.$$

Consequently, if $\mathbb{E}\sup_{t\ge 0}\|M_t\|^2 < \infty$, then

$$\mathbb{E}\sup_{t\ge 0}\|N_t\|^2 \le 4\,\mathbb{E}\sup_{t\ge 0}\|M_t\|^2\,.$$

Proof. First assume that M^τ and N^τ are square integrable. By the initial-coordinate domination, outside one null set

$$|\,\langle N_0,h\rangle\,| \le |\,\langle M_0,h\rangle\,| \qquad \text{for every } h\in H.$$

Taking $h = N_0 / \|N_0\|$ when $N_0 \neq 0$ gives $\|N_0\| \leq \|M_0\|$.

Characteristic domination of the continuous characteristics says that the bilinear form

$$\langle\!\langle M^c \rangle\!\rangle_\tau - \langle\!\langle N^c \rangle\!\rangle_\tau$$

is nonnegative. Therefore its trace is nonnegative, so

$$\operatorname{tr}\langle\!\langle N^c \rangle\!\rangle_\tau \leq \operatorname{tr}\langle\!\langle M^c \rangle\!\rangle_\tau.$$

The jump-compensator domination $\nu^N \leq \nu^M$ gives

$$\int_{[0,\tau]\times H} \|x\|^2 \, d\nu^N(t,x) \leq \int_{[0,\tau]\times H} \|x\|^2 \, d\nu^M(t,x).$$

Applying the Hilbert square identity to N^τ and M^τ, and comparing the three terms, yields

$$\mathbb{E}\,\|N_\tau\|^2 \leq \mathbb{E}\,\|M_\tau\|^2 .$$

For general local martingales, apply this argument to $\tau \wedge \rho_k$, where (ρ_k) is a common localizing sequence for which both stopped martingales are square integrable. Characteristic domination is preserved by stopping. If $\mathbb{E}\,\|M_\tau\|^2 = \infty$, there is nothing to prove. Otherwise, optional sampling for the submartingale $\|M_t\|^2$ gives $\mathbb{E}\,\|M_{\tau\wedge\rho_k}\|^2 \leq \mathbb{E}\,\|M_\tau\|^2$, while Fatou's lemma gives

$$\mathbb{E}\,\|N_\tau\|^2 \leq \liminf_k \mathbb{E}\,\|N_{\tau\wedge\rho_k}\|^2 \leq \mathbb{E}\,\|M_\tau\|^2 .$$

Finally, for each deterministic T, Doob's L^2 maximal inequality for the submartingale $\|N_t\|$ gives

$$\mathbb{E} \sup_{0\leq t\leq T} \|N_t\|^2 \leq 4\,\mathbb{E}\,\|N_T\|^2 \leq 4\,\mathbb{E}\,\|M_T\|^2 \leq 4\,\mathbb{E} \sup_{0\leq t\leq T} \|M_t\|^2 .$$

Letting $T \to \infty$ proves the stated strong estimate. □

Corollary 1 (Discrete aggregate domination in Hilbert space at $p = 2$)**.** *Let $(\xi_n)_{n=1}^N$ and $(\eta_n)_{n=1}^N$ be independent mean-zero H-valued random variables. If*

$$\sum_{n=1}^N \mathbb{P}(\xi_n \in B) \leq \sum_{n=1}^N \mathbb{P}(\eta_n \in B) \qquad \text{for every Borel } B \subset H \setminus \{0\},$$

then

$$\mathbb{E}\left\|\sum_{n=1}^N \xi_n\right\|^2 \leq \mathbb{E}\left\|\sum_{n=1}^N \eta_n\right\|^2 .$$

Proof. Independence and the mean-zero assumption give orthogonality in $L^2(H)$:

$$\mathbb{E}\left\|\sum_{n=1}^N \xi_n\right\|^2 = \sum_{n=1}^N \mathbb{E}\,\|\xi_n\|^2 .$$

The aggregate domination hypothesis, integrated against the nonnegative function $x \mapsto \|x\|^2$, gives

$$\sum_{n=1}^N \mathbb{E}\,\|\xi_n\|^2 \leq \sum_{n=1}^N \mathbb{E}\,\|\eta_n\|^2 .$$

The same orthogonality identity for (η_n) completes the proof. □

Verification Notes

The proof is insensitive to accessible versus quasi-left-continuous jumps: both jump types enter the Hilbert L^2 identity through the same scalar quantity $\int \|x\|^2 \, d\nu$. Thus the argument addresses precisely the general-local-martingale issue at $p = 2$, not only the quasi-left-continuous case already handled in the source paper.

The result is compatible with the previous c_0 counterexample. That counterexample exploits the failure of Hilbert square additivity in sup norm. In Hilbert space, grouping the same aggregate jump mass among independent variables cannot change the terminal second moment.

Novelty and Literature Check

The run indexes were searched for `1907.11588`, `characteristically dominated`, `aggregate domination`, `Hilbert`, and `L2`; no prior run packet for this subcase was found. Web searches for the exact problem and nearby phrases found the source paper, Montgomery-Smith–Pruss comparison inequalities [2], and Rosenthal/Pinelis moment tools, but no exact recorded answer to the Hilbert $p = 2$ general-local-martingale subcase.

This packet should be read as a new observation for the run rather than as a claim that the Hilbert L^2 estimate is technically deep: the proof is the standard Hilbert martingale square identity applied to the source paper's characteristic-domination data.

Limitations and Next Routes

This does not settle the original UMD-space problem for all p, nor even the Hilbert-space problem for $p \neq 2$. For $p > 2$, the plausible route is to combine Hilbert-valued martingale Rosenthal or Bichteler-Jacod inequalities with domination of the second and p-th jump characteristic terms. That route was not promoted here because this packet keeps only the completely self-contained L^2 argument.

Human Review Recommendation

Review the passage from bilinear-form domination to trace domination and the localization step. If those check out, this should be accepted as a positive Hilbert-valued $p = 2$ subcase of the general characteristically dominated local martingale problem.

References

[1] I. S. Yaroslavtsev, *Local characteristics and tangency of vector-valued martingales*, arXiv:1907.11588, 2019.

[2] S. J. Montgomery-Smith and A. R. Pruss, *A comparison inequality for sums of independent random variables*, J. Math. Anal. Appl. 254 (2001), no. 1, 35–42; arXiv:math/9811124.